

Computational Neuroscience — Module 01: Systems

— Study Questions

1. What defines the brain as a dynamical system? What are the "states," "rules," and "trajectories" in neural dynamics?
2. How does a single neuron perform computation? Describe the input-integration-output process. Why is a single neuron not sufficient for cognition?
3. What are neural oscillations? Why do populations of neurons naturally produce rhythmic firing patterns? What creates these rhythms?
4. Compare alpha, beta, gamma, and theta oscillations. What cognitive function is each associated with? Why does the brain need multiple frequency bands?
5. What is an attractor? How does the "ball rolling into a valley" metaphor help explain stable neural patterns?
6. Compare fixed-point, limit-cycle, and strange attractors. Give a neural example of each. Which is most common in normal brain function?
7. How does memory recall relate to attractor dynamics? When you see part of a friend's face, how does attractor dynamics explain why you can recognize the whole face?
8. What is a Hopfield network? How does it implement associative memory as a dynamical system with attractor states?
9. How does the Markov Blanket concept apply to a single neuron? What are its sensory states (dendrites), active states (axon), and internal states (membrane potential)?
10. How does the Markov Blanket scale from neuron level to brain region level? Why is this hierarchical structure important for Active Inference?
11. What is self-organization? How do excitatory and inhibitory interactions create organized patterns without any central "conductor"?
12. How might neural dynamics relate to consciousness? Does consciousness correspond to a specific type of attractor? What would the dynamical signature of consciousness look like?
13. How do epileptic seizures relate to attractor dynamics? Is a seizure a pathological attractor — a stable pattern of neural activity that the brain gets "stuck" in?
14. Compare the brain to other dynamical systems (weather, the stock market, a flock of birds). What dynamics do they share, and where does the brain differ?

15. Critically evaluate: Can the entire brain be adequately modeled as a dynamical system, or are there aspects of brain function that dynamical systems theory misses?
16. What is the difference between a system's behavior being "complicated" versus "complex"? A jet engine is complicated; a flock of starlings is complex. What property distinguishes them, and where does the brain fall on this spectrum?
17. How do feedback loops create emergent behavior? Describe a positive feedback loop and a negative feedback loop in the nervous system, and explain how their interaction produces stable yet flexible behavior.
18. What is the concept of "degeneracy" in neural systems — the idea that many different neural configurations can produce the same function? Why is this important for understanding brain resilience after injury?
19. Explain how the concept of a "phase transition" applies to neural systems. What might a phase transition in brain dynamics look like, and how does this relate to phenomena like sleep-wake transitions or seizure onset?
20. How does the brain implement multiscale organization — from individual ion channels to neural circuits to large-scale networks? Why is it that phenomena at one scale (e.g., a single synapse changing strength) can have consequences at a completely different scale (e.g., memory formation)?